

Micro Syllabus for Statistics (B.Sc. CSIT) program

Tribhuvan University
Institute of Science & Technology
(IOST)

Level: B.Sc.

Course Title: Statistics I

Course Code: STA 164

Nature of the Course: Theory and Practical

Full Marks: 60 + 20 + 20

Pass Marks: 24 + 8 + 8

Credit Hours: 3

Course objectives:

To impart the knowledge of descriptive statistics, correlation, regression, sampling, theoretical as well as the applied knowledge of probability and some probability distributions

1. Introduction

[4]

Basic concept of statistics: Definitions, descriptive and inferential statistics, limitations of statistics; Application of Statistics in the field of Computer Science & Information technology; Scales of measurement: nominal, ordinal, interval and ratio scale; Variables: qualitative, quantitative, discrete and continuous variables; Types of Data: primary data, secondary data and their sources, cross-sectional data, time series data, failure time data, panel data; Notion of a statistical population: finite population, infinite population, homogeneous population and heterogeneous population.

2. Descriptive statistics

[6]

Measures of central tendency: mean, combined arithmetic mean, weighted mean, median, mode, partition values; Measures of dispersion: absolute and relative measures of dispersion, range, quartile deviation, mean deviation, standard deviation, combined standard deviation, coefficient of variation; Measure of skewness: Bowley's coefficient of skewness, Karl Pearson's coefficient of skewness; Measure of kurtosis: leptokurtic, mesokurtic and platykurtic frequency distributions, measures of kurtosis using partition values; Moments: raw and central moments, measures of skewness and kurtosis based on moments; stem and leaf display; five number summary; box plot; normal probability plot
Problems and illustrative examples related to computer Science and IT

3. Introduction to Probability

[8]

Concepts of probability; Deterministic and random experiments; Basic terminology: trial and event, outcome, sample space, equally likely, mutually exclusive, exhaustive and favourable cases, sure and impossible events, independent and dependent events; Definitions of probability: mathematical (classical), statistical (relative frequency) and subjective definition; Laws of probability: Additive and multiplicative theorems; Conditional probability; Bayes theorem; prior and posterior probabilities.

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4. Sampling

[3]

Definitions of population, sample survey vs. census survey, sample, sampling unit, sampling frame, sampling error and non-sampling error; Steps in sampling; Fundamental characteristics, advantages and disadvantages of sampling; Types of sampling: Probability (simple, stratified, systematic, cluster and multistage); non probability sampling (convenience, purposive, quota, snowball).

5. Random Variables and Mathematical Expectation

[5]

Concept of a random variable; Types of random variables: Discrete and continuous random variables; Probability distribution of a random variable: probability mass function and probability density function, distribution function and its properties; Mathematical expectation of a random variable (discrete and continuous); Properties of mathematical expectation of random variables; Addition and multiplicative theorems of expectation; Concept of conditional expectations; variance, standard deviation and covariance; Properties of covariance and variance(without derivation)

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6. Probability Distributions

[12]

Concept of Probability distribution function, Joint probability distribution of two random variables: Joint probability mass function and density functions, marginal probability mass and density function; Conditional distribution functions; statistical independence; Discrete distributions: Bernoulli trial, Binomial and Poisson distributions, their mass functions, mean and standard deviation of their distribution, fitting of binomial and Poisson distribution(without derivation); Continuous distribution: Normal distributions, its probability density functions, measurement of area under the normal curve; Standardization of normal distribution; Normal distribution as an approximation of Binomial and Poisson distribution (without derivation); Exponential, Gamma distribution, measure characteristics(without derivation) and their application in relevant areas.

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7. Correlation and Linear Regression

[7]

Bivariate data; Bivariate frequency distribution; Correlation between two variables; Positive correlation; Negative correlation; Scatter diagram to explore the type of correlation; Karl Pearson's coefficient of correlation (r): Definition, computation for grouped and ungrouped data and interpretation; Properties of correlation coefficient; Spearman's rank correlation including tied cases; Regression Analysis: Concept of regression, lines of regression, fitting of lines of regression by the least squares method, interpretation of slope and intercept, concept of linearity; properties of regression coefficient; explained and unexplained variation, residual analysis; coefficient of determination.

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Practical (Computational Statistics):**[15]**

Practical problems to be covered in the Computerized Statistics laboratory

Practical problems

S. No.	Title of the practical problems (Using any statistical software such as Microsoft Excel, SPSS, STATA etc. whichever convenient).	No. of practical problems
1	Computation of measures of central tendency (ungrouped and grouped data) Use of an appropriate measure and interpretation of results and computation of partition Values	1
2	Computation measures of dispersion (ungrouped and grouped data) and computation of coefficient of variation.	1
3	Measures of skewness and kurtosis using method of moments, Measures of Skewness using Box and whisker plot, normal probability plot	2
4	Scatter diagram, correlation coefficient (ungrouped data) and interpretation. Compute manually and check with computer output.	1
5	Fitting of lines of regression (Results to be verified with computer output)	1
6	Fitting of lines of regression and computation of correlation coefficient, Mean residual sum of squares, residual plots.	1
7	Conditional probability and Bayes theorem	3
8	Obtaining descriptive statistics of probability distributions	2
9	Fitting probability distributions in real data (Binomial, Poisson and Normal)	3
	Total number of practical problems	15

Text Books:

1. Michael Baron (2013). Probability and Statistics for Computer Scientists. 2nd Ed., CRC Press, Taylor & Francis Group, A Chapman & Hall Book.
2. Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, & Keying Ye(2012). Probability & Statistics for Engineers & Scientists. 9th Ed., Printice Hall.

Reference Books:

1. Douglas C. Montgomery & George C. Ranger (2003). Applied Statistics and Probability for Engineers. 3rd Ed., John Willey and Sons, Inc.
2. Richard A. Johnson (2001). Probability and Statistics for Engineers. 6th Ed., Pearson Education, India